

Homework 4

Diya Jain

1. $x \equiv 1 \pmod{2}$

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

This congruence class has infinitely many elements, hence there are infinitely many solutions in $\mathbb{Z}$.

$$\text{class modulo} = 2 \times 3 \times 5 = 30$$

$$A_1 = 3 \times 5 = 15$$

$$x_1 = 1$$

$$a_1 = 2$$

$$A_2 = 2 \times 5 = 10$$

$$x_2 = 2$$

$$a_2 = 3$$

$$A_3 = 2 \times 3 = 6$$

$$x_3 = 3$$

$$a_3 = 5$$

$$y_1 = A_1^{-1} \pmod{2} = 1$$

$$y_2 = A_2^{-1} \pmod{3} = 1$$

$$y_3 = A_3^{-1} \pmod{5} = 1$$

$$x = x_1 y_1 A_1 + x_2 y_2 A_2 + x_3 y_3 A_3$$

$$= (1)(1)(15) + (2)(1)(10) + (3)(1)(6)$$

$$= 15 + 20 + 18$$

$$= 53$$

Hence, all solutions are given by $53 + k \cdot 30$ for $k \in \mathbb{Z}$.

After simplifying we get the solutions are of the form $23 + 30 \cdot k$ for $k \in \mathbb{Z}$.

Solutions in $\{0, \dots, 769\}$:

$$x = 30n + 23 \quad \text{where } 0 \leq 30n + 23 \leq 769$$

Solving $30n + 23 \leq 769$ gives:

$$n \leq 24.866 \Rightarrow n = 0, 1, 2, \dots, 24$$

which gives us 25 solutions.

2. $\phi(69069)$

3	69069
7	23023
11	3289
13	299
23	23
	1

$$\phi(3' \cdot 7' \cdot 11' \cdot 13' \cdot 23') = \phi(3') \times \phi(7') \times \phi(11') \times \phi(13') \times \phi(23')$$

$$= 3^0(2) \times 7^0(6) \times 11^0(10) \times 13^0(12) \times 23^0(22)$$

$$= (2)(6)(10)(12)(22)$$

$$= 31680 \text{ ans.}$$